

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 1.4: ENERGY, WORK, POWER AND MACHINES

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the whole unit. Read the driving question: Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails, and where does the energy go when friction makes the nut hot? Answer each section in the space below its prompt. Write in full sentences, show every step of any calculation with units, and use the words and formulae you learned in Lessons 1 to 8. Link every answer back to the mechanic at Gikomba market. Use $g = 10 \text{ m/s}^2$. Work on your own.

Section 1: The Mechanic at Gikomba and Energy of Motion and Position

A jua kali mechanic in Gikomba market, Nairobi, fails to loosen a matatu wheel nut with a short spanner. He then succeeds easily with the spanner extended by a metal pipe. An apprentice notices the nut feels warm after the failed attempts.

- (a) State the two puzzles (mysteries) this scene raised at the start of the unit. (2 marks)
- (b) Write the formulae for kinetic energy and gravitational potential energy and name the unit of energy. (3 marks)
- (c) A 70 kg matatu tout runs at 4 m/s. Calculate his kinetic energy. (2 marks)
- (d) A 5 kg bag of maize is lifted 2 m onto a grain-store shelf in Kitale. Calculate its gravitational potential energy. (2 marks)
- (e) If the tout doubled his speed to 8 m/s, what would happen to his kinetic energy? Explain using the formula. (2 marks)

Write your answer here:

Section 2: Work, Force and Displacement

In Lesson 3 you looked at the data from the mechanic's two attempts, using $W = Fd \cos\theta$ as your model of work. In each row, F is the force applied and d is the displacement in the direction of that force ($\theta = 0^\circ$).

Spanner	Force applied (N)	Displacement (m)
Short	250	0
Long	80	0.45

(a) Write the formula for work, including the angle. (2 marks)

(b) Use the formula to calculate the work for each row of the table. Show your steps. (4 marks)

(c) The mechanic pushed hard with the short spanner and got tired. Explain, using the formula, why a scientist says no work was done in the short-spanner row. (2 marks)

(d) Name the three kinds of work (by sign) that depend on the angle between force and displacement and give the angle for each. Give one example of each from your lessons. (3 marks)

Write your answer here:

Section 3: Where Does the Energy Go?

The apprentice felt that the wheel nut was warm after the mechanic's attempts with the short spanner.

- (a) Explain why the nut became warm even though it did not move. Name the energy change that took place. (3 marks)
- (b) In Lesson 4 you pulled a wooden block over smooth tile and over sandpaper. Write the formula for work done against friction and explain what you found about the warmth of the two surfaces and why. (3 marks)
- (c) Write an equation showing how total work input, useful work output and work done against friction are related. (1 mark)
- (d) A classmate says: "The heat means energy was destroyed." Use the Law of Conservation of Energy to explain why this is wrong. (3 marks)

Write your answer here:

Section 4: How the Long Spanner Helps: Machines, MA, VR and Efficiency

The spanner and its pipe extension work as a lever.

- (a) Write the formulae for mechanical advantage (MA), velocity ratio (VR) and efficiency. (3 marks)**
- (b) Explain how the long spanner changes the effort needed to loosen the nut. Say what happens to the distance the mechanic's hand moves compared with the distance the load moves. (3 marks)**
- (c) A classmate says: "The long spanner gives the mechanic extra energy." Use the idea of work in and work out to say whether you agree. (3 marks)**
- (d) Explain why the efficiency of any real machine, including the spanner, is always below 100%. (2 marks)**

Write your answer here:

Section 5: The Full Story and Power

Now put the whole unit together to answer the driving question: Why can the jua kali mechanic loosen the wheel nut effortlessly with a long spanner when a short one fails, and where does the energy go when friction makes the nut hot?

Write a clear explanation of about 6 to 8 sentences. Use at least four of these terms correctly: work, mechanical advantage, velocity ratio, friction, thermal energy, efficiency, conservation of energy, power.

Then answer: Two mechanics loosen the same nut and do the same work, but one takes less time. Which one has the greater power? Write the formula for power and its unit. (Total: 8 marks for the explanation, 3 marks for power)

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculations and use of formulae (KE, PE, work)	All calculations are correct, with formulae stated, steps shown and units given. The meaning of $d = 0$ and the effect of doubling speed are explained correctly.	Most calculations are correct with formulae and units. There may be a small slip in a step or a unit, or the explanation of zero work or of doubling speed is incomplete.	Formulae are missing or wrongly used. Calculations have several errors or no units, and zero work is not explained.
Explaining energy transformation and dissipation (friction and heat)	Clearly explains that friction changes muscular energy into thermal energy, uses $W_{in} = W_{out} + W_{friction}$, and correctly argues that energy is conserved and not destroyed.	States that friction produces heat and that energy is conserved, but the link to work against friction or the equation is thin or partly missing.	Says energy is lost or destroyed, or gives no mechanism for the warm nut.
Explaining machines, MA, VR and efficiency	Explains that the longer effort arm raises MA and that force is traded for distance. States that no machine creates energy and that efficiency is below 100% because friction makes work out less than work in.	Explains the lever advantage correctly but is vague about the force-distance trade or about why efficiency is below 100%.	Suggests the machine gives extra energy, or cannot state the MA, VR or efficiency formulae.
Integrated explanation of the phenomenon and use of vocabulary	A coherent, linked explanation of both puzzles that uses at least four unit terms correctly, with power distinguished as	A mostly connected explanation using four terms, with minor misuse, or a power answer that is correct but incomplete.	A list of facts that does not link to the mechanic, uses fewer than four terms, or confuses power with work.

	rate of working ($P = W/t$, watts).		
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